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### Micro device integration into system substrate

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#### Abstract

This disclosure is related to post processing steps for integrating of micro devices into system (receiver) substrate or improving the performance of the micro devices after transfer. Post processing steps for additional structure such as reflective layers, fillers, black matrix or other layers may be used to improve the out coupling or confining of the generated LED light. In another example, dielectric and metallic layers may be used to integrate an electro-optical thin film device into the system substrate with the transferred micro devices. In another example, color conversion layers are integrated into the system substrate to create different output from the micro devices.

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#### **Background/Summary**

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/912,049, filed Jun. 25, 2020 (3USD3), now allowed, which is a division of U.S. application Ser. No. 15/653,120, filed Jul. 18, 2017 (3USP3), now U.S. Pat. No. 10,700,120, issued Jun. 30, 2020, which is a continuation-in-part of, and claims priority from, U.S. application Ser. No. 15/060,942, filed Mar. 4, 2016 (3USP1), now U.S. Pat. No. 10,134,803, issued Nov. 20, 2018, which is a continuation-in-part and claims priority from U.S. application Ser. No. 15/002,662, filed Jan. 21, 2016 (3USPT), abandoned, which claims priority from Canadian Application No. 2,890,398, filed May 4, 2015 (129CAPL), Canadian Application No. 2,883,914, filed Mar. 4, 2015 (128CAPL), Canadian Application No. 2,880,718, filed Jan. 28, 2015 (127CAPL), Canadian Application No. 2,879,465, filed Jan. 23, 2015 (125CAPL), and Canadian Application No. 2,879,627, filed Jan. 23, 2015 (99CAPL), each of which is hereby incorporated by reference herein in its entirety. This application also claims priority to Canadian Application No. 2,936,473, filed Jul. 19, 2016 (14CAPL), which is hereby incorporated by reference herein in its entirety.

## TECHNICAL FIELD

(1) The present disclosure relates to the integration of transferred micro device systems onto a receiver substrate. More specifically, the present disclosure relates to the post processing steps for enhancing the performance of micro devices after transferring into a receiver substrate including the development of optical structure, the integration of electro-optical thin film devices, the addition of color conversion layers, and the proper patterning of devices on a donor substrate.

## BACKGROUND

(2) An object of the present invention is to overcome the shortcomings of the prior art by providing a light emitting micro device and a thin film electro-optical light emitting device integrated on a same receiver substrate.

## SUMMARY OF THE INVENTION

(3) Accordingly, the present invention relates to an integrated optical system comprising a plurality of pixels, each pixel comprising:

- (4) a receiver substrate;
- (5) a light emitting micro device integrated on the receiver substrate;
- (6) a planarization or bank region surrounding the micro device; and
- (7) a thin-film light emitting electro-optical device, at least a portion of which is mounted on the planarization or bank region.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention will be described in greater detail with reference to the accompanying drawings which represent preferred embodiments thereof, wherein:

- (2) FIG. 1 shows a receiver substrate with contact pads, and an array of transferred micro-devices attached to the receiver substrate.
- (3) FIG. 2A shows a receiver substrate with contact pads, an array of transferred micro-devices attached to the receiver substrate, and conformal dielectric and reflective layers on top.
- (4) FIG. 2B shows a receiver substrate with contact pads, an array of transferred micro-devices attached to the receiver substrate, and patterned conformal dielectric and reflective layers.
- (5) FIG. 2C shows a receiver substrate with contact pads, an array of transferred micro-devices attached to the receiver substrate, patterned conformal dielectric and reflective layers, and a black matrix layer formed between adjacent micro-devices.
- (6) FIG. 3A shows a receiver substrate with contact pads, an array of transferred micro-devices

attached to the receiver substrate, patterned conformal dielectric and reflective layers, a black matrix layer, and a transparent conductive layer deposited on the substrate.

(7) FIG. 3B shows a receiver substrate with an integrated array of transferred micro-devices attached to the receiver substrate and optical reflective components for light outcoupling enhancement.

(8) FIG. 3C shows a receiver substrate with an integrated array of transferred micro-devices attached to the receiver substrate and concave contact pads for light outcoupling enhancement.

(9) FIG. 3D shows a receiver substrate with an integrated array of transferred micro-devices attached to the receiver substrate in a bottom emission configuration.

(10) FIG. 3E shows a receiver substrate with an integrated array of transferred micro-devices attached to the receiver substrate.

(11) FIG. 4A shows a receiver substrate with transferred micro-devices, a conformal dielectric layer, and a connected reflective layer.

(12) FIG. 4B shows a receiver substrate with transferred micro-devices, conformal dielectric layer, connected reflective layer, and a transparent conductive layer deposited on the substrate.

(13) FIG. 5 shows a receiver substrate with transferred micro-devices and a patterned filler which defines the pixels (or sub-pixels).

(14) FIG. 6A shows a pixelated filler structure covering all sub-pixels in at least one pixel (for example covering both sub-pixels for a pixel made of two sub-pixels).

(15) FIG. 6B shows a pixel made of two sub-pixels, a filler layer which is patterned to define the pixel, and patterned conformal dielectric and reflective layers around the pixel.

(16) FIG. 6C shows a pixel made of two sub-pixels, a filler layer which is patterned to define the pixel, patterned conformal dielectric and reflective layers around the pixel, and a black matrix layer wrapped around the pixel.

(17) FIG. 6D shows a pixel made of two sub-pixels, a filler layer which is patterned to define the pixel, patterned conformal dielectric and reflective layers around the pixel, a black matrix layer wrapped around the pixel, and a transparent conductive layer deposited on the substrate.

(18) FIG. 6E shows a pixel made of two sub-pixels with reflective optical components on the receiver substrate for better light outcoupling.

(19) FIG. 6F shows a pixel made of two sub-pixels with concave contact pads on the receiver substrate.

(20) FIG. 6G shows a pixel made of two sub-pixels with a bottom emission configuration.

(21) FIG. 6H shows a pixel made of two sub-pixels with a bottom emission configuration, a common top electrode, and side reflectors.

(22) FIG. 7 shows a receiver substrate with two contact pads.

(23) FIG. 8 shows a receiver substrate with a transferred micro device bonded to one of the contact pads.

(24) FIG. 9 shows the integration of a transferred micro-device with an electro-optical thin film device in a hybrid structure.

(25) FIG. 10 shows another example of an integration of a transferred micro-device with an electro-optical thin film device in a hybrid structure.

(26) FIG. 11 shows an example of the integration of a transferred micro-device with an electro-optical thin film device in a hybrid structure with a common top electrode.

(27) FIG. 12 shows an embodiment for the integration of a transferred micro-device with an electro-optical thin film device in a dual surface hybrid structure with both top and bottom transparent electrodes.

(28) FIG. 13A shows another embodiment for a system substrate and an integrated micro device with thin film electro-optical device.

(29) FIG. 13B shows another embodiment of a system substrate and an integrated micro device with a thin film electro-optical device.

(30) FIG. 14A shows an embodiment of a system substrate and an integrated micro device with two thin film electro-optical devices.

(31) FIG. 14B shows an example of a system substrate and an integrated micro device with two thin film electro-optical devices and a reflective layer on the receiver substrate.

(32) FIG. 14C and FIG. 14D show example of micro-LED and optoelectronic device where the light passes through the optoelectronic device.

(33) FIG. 14E, FIG. 14F, and FIG. 14G show another examples of micro-LED integrated with optoelectronic device on system substrate.

(34) FIG. 14H and FIG. 14I show another example of micro-LED integrated with optoelectronic device and color filter on system substrate.

(35) FIG. 14J shows another example of micro-LED integrated with optoelectronic device and color filter on system substrate.

(36) FIG. 15 illustrates a cross section of a system substrate and a micro device substrate.

(37) FIG. 16 shows the alignment step for a system substrate and a micro device substrate in a transfer process.

(38) FIG. 17 shows the bonding step for a system substrate and a micro device substrate in a transfer process.

(39) FIG. 18 shows the micro device substrate removal step for a system substrate and a micro device substrate in a transfer process.

(40) FIG. 19 shows the sacrificial layer removal step for a system substrate and a micro device substrate in a transfer process.

(41) FIG. 20 shows the common electrode formation step for a system substrate and a micro device substrate in a transfer process.

(42) FIG. 21 is a cross section of a micro device substrate with a filler layer(s).

(43) FIG. 22 is a cross section of a micro device substrate covered with a support layer.

(44) FIG. 23 shows the micro device substrate removal step for a micro device substrate in a transfer process.

(45) FIG. 24A and FIG. 24B show the sacrificial layer removal step for a micro device substrate in a transfer process. A system substrate with contact pads is shown as well.

(46) FIG. 25 shows the bonding step for a system substrate and a micro device substrate in a transfer process.

(47) FIG. 26A and FIG. 26B show the supporting layer removal step for a micro device substrate in a transfer process. A system substrate with contact pads and transferred micro devices is shown as well.

(48) FIG. 27 is a cross section of a micro device substrate covered with a filler layer.

(49) FIG. 28A and FIG. 28B are cross sections of a micro device substrate with via holes in the substrate and the sacrificial layer.

(50) FIG. 29 is a cross section of a micro device substrate with via holes in the substrate and the sacrificial layer covered by an insulating layer.

(51) FIG. 30 is a cross section of a micro device substrate with a conductive layer filled via holes in the substrate and the sacrificial layer.

(52) FIG. 31 is a cross section of a micro device substrate with a common top electrode.

(53) FIG. 32 is a cross section of an integrated system substrate with a common top electrode.

(54) FIG. 33A shows a two dimensional arrangement of micro devices in a donor substrate.

(55) FIG. 33B is a cross section of a system substrate and a micro device substrate.

(56) FIG. 34 is a cross section of a bonded system substrate and micro device substrate.

(57) FIG. 35 shows the laser lift-off step for a micro device substrate in a transfer process.

(58) FIG. 36 is a cross section of a system substrate and a micro device substrate after the selective transfer process.

(59) FIG. 37 shows an integrated system substrate with a common top electrode.

- (60) FIGS. 38A and 38B are cross sections of a micro device substrate with micro devices having different heights.
- (61) FIG. 39 is a cross section of a micro device substrate with a filler layer.
- (62) FIG. 40 shows the alignment step for a system substrate with grip mechanisms and a micro device substrate in a transfer process.
- (63) FIG. 41A shows a two dimensional arrangement of micro devices in a donor substrate.
- (64) FIG. 41B is a cross section of a system substrate and a micro device substrate with different pitches.
- (65) FIG. 42 shows the selective micro device transfer process for a system substrate and a micro device substrate with different pitches.
- (66) FIG. 43 is a cross section of a system substrate and a micro device substrate with different pitches.
- (67) FIG. 44 shows the selective micro device transfer process for a system substrate and a micro device substrate with different pitches.
- (68) FIG. 45 shows an integrated micro device substrate.
- (69) FIG. 46A and FIG. 46B show the transfer process of micro devices to a system substrate with a planarization layer, a common top electrode, bank structures, and color conversion elements.
- (70) FIG. 47 shows a structure with color conversion for defining the color of pixels.
- (71) FIG. 48 shows a structure with conformal common electrode and color conversion separated by a bank layer.
- (72) FIG. 49 shows a structure with conformal color conversion separated by a bank layer.
- (73) FIG. 50 shows a structure with color conversion elements on the common electrode without the bank layer.
- (74) FIG. 51 shows a structure with conformal common electrode and color conversion.
- (75) FIG. 52 shows a structure with conformal color conversion elements formed directly on the micro devices.
- (76) FIG. 53A and FIG. 53B show a structure with color conversion for defining pixel color, a planarization layer, and a common transparent electrode.
- (77) FIG. 54A and FIG. 54B show a structure with color conversion for defining pixel color and a separate substrate for encapsulation.
- (78) FIG. 55A, FIG. 55B, FIG. 55C and FIG. 55D show a structure with color conversion for defining pixel color while the pixelation is done with current confinement methods.

#### DETAILED DESCRIPTION

- (79) While the present teachings are described in conjunction with various embodiments and examples, it is not intended that the present teachings be limited to such embodiments. On the contrary, the present teachings encompass various alternatives and equivalents, as will be appreciated by those of skill in the art.
- (80) The process of developing a system based on micro devices comprises pre-processing the devices on a donor substrate (or a temporary substrate), transferring the micro devices from the donor to the receiver substrate, and post processing to enable device functionality. The pre-processing step may include patterning and adding bonding elements. The transfer process may involve bonding of a pre-selected array of micro devices to the receiver substrate followed by removing the donor substrate. Several different selective transfer processes have been already been developed for micro devices. After the integration of the micro devices into the receiving substrate, additional post processes may be performed to make required functional connections.
- (81) In this disclosure, emissive device is used to describe different integration and post processing methods. However, it is clear for one skill in the art that other devices such as sensors can be used in these embodiments. For example, in case of sensor micro devices, the optical path will be similar to emissive micro devices but in reverse direction.
- (82) Some embodiments of this disclosure are related to post-processing steps for improving the



performance of the micro devices. For example, in some embodiments, the micro device array may comprise micro light emitting diodes (LEDs), Organic LEDs, sensors, solid state devices, integrated circuits, (micro-electro-mechanical systems) MEMS, and/or other electronic components. The receiving substrate may be, but is not limited to, a printed circuit board (PCB), thin film transistor backplane, integrated circuit substrate, or, in one case of optical micro devices such as LEDs, a component of a display, for example a driving circuitry backplane. In these embodiments, in addition to interconnecting the micro devices, post processing steps for additional structure such as reflective layers, fillers, black matrix or other layers may be used to improve the out coupling of the generated LED light. In another example, dielectric and metallic layers may be used to integrate an electro-optical thin film device into the system substrate with the transferred micro devices.

(83) In one embodiment, the active area of the pixel (or sub-pixel) is extended to be larger than the micro device by using fillers (or dielectric). Here, the filler is patterned to define the pixel's active area (the active area is the area that emits light or is absorbing input light). In another embodiment, reflective layers are used to confine the light within the active area.

(84) In one embodiment, the reflective layer may be one of the micro device electrodes.

(85) In another embodiment, the active area may comprise a few sub-pixels or pixels.

(86) The active area can be larger, smaller, or the same size as the pixel (sub-pixel) area.

(87) In another embodiment, thin film electro-optical devices are deposited into the receiver substrate after the micro devices are integrated into the receiver substrate.

(88) In one embodiment, an optical path is developed for the micro device to emit (or absorb) light through all or some layers of the electro-optical device.

(89) In another embodiment, the optical path for the micro device is not through all or some layers of the optoelectronic device.

(90) In one embodiment, the optoelectronic device is a thin film device.

(91) In another embodiment, the electrode of the electro-optical device is used to define the active area of the pixel (or sub-pixel).

(92) In another embodiment, at least one of the electro-optical device electrodes is shared with the micro-device electrode.

(93) In one embodiment color conversion material covers the surface and surrounds partially (or fully) the body of the micro device.

(94) In one embodiment, the bank structure separates the color conversion materials.

(95) In another embodiment color conversion material covers the surface (and/or partially or fully the body of) the active area.

(96) In one embodiment, the micro devices on donor substrate are patterned to match the array structure in the receiver (system) substrate. In this case, all the devices in part (or all) of the donor substrate are transferred to the receiver substrate.

(97) In another embodiment, VIAs are created in the donor substrate to couple the micro devices on the donor substrate with the receiver substrate.

(98) In another embodiment, the donor substrate has more than one micro device types and at least in one direction the pattern of the micro device types on the donor substrate matches partially or fully the pattern of the corresponding areas (or pads) on the system substrate.

(99) In another embodiment, the donor substrate has more than one micro device types and at least in one direction the pitch between different micro devices in donor substrate is a multiple of the pitch of the corresponding area (or pads) on the system substrate.

(100) In another embodiment, the donor substrate has more than one micro device type. At least in one direction, the pitch between two different micro devices matches the pitch of the corresponding areas (or pads) on the receiver (or system) substrate.

(101) In one embodiment, the pattern of different micro device types on the donor substrate creates a two dimensional array of each type where the pitch between each array of different types matches

the pitch of the corresponding areas on the system substrate.

(102) In another embodiment, the pattern of different micro device types on the donor substrate create a one dimensional array where the pitch of the arrays matches the pitch of the corresponding areas (or pads) on the system substrate.

(103) FIG. 1 shows a receiver substrate **100**, contact pads **101a** and **101b**, and micro devices **102a** and **102b**, being in an array attached to the receiver substrate **100**. The contact pads **101a** and **101b** onto which the micro devices **102a** and **102b** have been transferred, are located in an array parallel to and mounted on the receiver substrate **100**. The micro devices **102a** and **102b** are transferred from a donor substrate and bonded to the contact pads **101a** and **101b**. The micro devices **102a** and **102b** may be any micro device that may typically be manufactured in planar batches, including but not limited to LEDs, OLEDs, sensors, solid state devices, integrated circuits, MEMS, and/or other electronic components.

(104) As depicted in FIG. 2A, in one embodiment where the micro devices **102a** and **102b** are micro LEDs, a conformal dielectric layer **201** and a reflective layer **202** may be formed over the bonded micro LEDs **102a** and **102b**. In some embodiments, the conformal dielectric layer **201** is approximately 0.1-1  $\mu\text{m}$  thick, and may be deposited by any of a number of different thin film deposition techniques. The conformal dielectric layer **201** isolates the sidewalls of the micro LEDs **102a** and **102b** from the reflective layer **202**. In addition, the dielectric layer **201** passivates and protects the sidewalls of the micro LEDs **102a** and **102b**. The conformal dielectric layer **201** may also cover the top surface of the receiver substrate **100** between adjacent micro LED devices **102a** and **102b**. The conformal reflective layer **202** may be deposited over the dielectric layer **201**. The reflective layer **202** may be a single layer or made up of multiple layers. A variety of conductive materials may be used as the reflective layer **202**. In some embodiments, the conformal reflective layer **202** may be a metallic bilayer with a total thickness up to 0.5  $\mu\text{m}$ .

(105) Referring to FIG. 2B, the dielectric layer **201** and the reflective layer **202** may then be patterned by using for example lithographic patterning and etching to partially expose the top surface of micro LEDs **102a** and **102b**. In one embodiment where the micro LEDs **102a** and **102b** are integrated into a backplane of a display system, referring also to FIG. 2C, a black matrix **203** may be formed between adjacent micro LEDs **102a** and **102b**, and on the reflective layer **202** to reduce the reflection of the ambient light. In one example the black matrix **203** may be a layer of resins, such as polyimide or polyacrylic, in which particles of black pigment, such as carbon black, have been dispersed. In some embodiments, the thickness of the black matrix layer **203** may be 0.01-2  $\mu\text{m}$ . The black matrix layer **203** may be patterned and etched so as to expose the top surface of the micro LEDs **102a** and **102b** as shown in FIG. 2C. Optionally, the thickness of the black matrix **203** may be engineered to planarize the integrated substrate **100**. In another embodiment, a planarization layer, which may be made of organic insulating material, is formed and patterned to planarize the backplane substrate.

(106) Referring to FIG. 3A, a transparent conductive layer **301** may be conformally deposited over the substrate **100**, covering the black matrix **203** and the top surface of micro LEDs **102a** and **102b**. In some embodiments, the transparent electrode **301** may be 0.1-1  $\mu\text{m}$  thick layers of oxides, including but not limited to indium tin oxide (ITO) and Aluminum doped Zinc Oxide. In a case where the integrated assembly is a display structure, the transparent electrode **301** may be the common electrode of the micro LED devices **102a** and **102b**.

(107) Optionally, the reflective layer **202** may be used as a conductivity booster for the transparent electrode **301**. In this case, part of the reflective layer **202** may not be covered with the black matrix **203**, or other planarization layers, so that the transparent electrode layer **301** may connect to the reflective layer **202**.

(108) In another embodiment shown in FIG. 3B, a reflective or other type of optical component **302** may be formed on the substrate **100** to enhance out-coupling of light produced by the micro device **102a** and **102b**. The common contact **301** is transparent to allow light output through this

layer. These structures may be referred as top emission structures.

(109) Referring to FIG. 3C, the contact pads **101a** and **101b** may be formed to have a concave or other shaped structure to enhance the out-coupling of light produced by micro-devices **102a** and **102b**. The form of the contact pads **101a** and **101b** is not limited to the concave form and may have other forms depending on the micro device light emission characteristics.

(110) In an embodiment, referring to FIG. 3D, the structure is designed to output light from the substrate **100**. In these bottom emission structures, the substrate **100** may be transparent and the common electrode **303** is designed to be reflective for better light extraction.

(111) In another embodiment shown in FIG. 3E, the reflective layer **202** may be extended to cover the micro devices **102a** and **102b**, and act as the common top electrode as well.

(112) Referring to FIG. 4A, in another embodiment, the dielectric layer **201** may be deposited and patterned before forming the reflective layer **202**, which may allow a direct contact between micro LEDs **102a** and **102b**, and the reflective layer **202**. Accordingly the reflective layer **202** may be used as a common top contact for the micro devices **102a** and **102b**. The black matrix **203** or alternatively a planarization layer may be used.

(113) Referring to FIG. 4B, in other embodiments, a common transparent electrode **301** or/and other optical layers may be deposited on top of the substrate **100** to enhance conductivity and/or light out coupling.

(114) One of the main challenges with micro optoelectronic devices is the empty space between adjacent micro devices **102a** and **102b**. Display systems with this structural characteristic may create an image artifact called the “screen door effect”. In one embodiment, the micro device sizes may be optically extended to be the same or larger than the micro device size. In one embodiment shown in FIG. 5, after transferring the array of micro devices **102a** and **102b** from the donor to the receiver substrate **100**, a transparent filler **501** may be deposited and patterned to define the pixel (or sub-pixel). In one example, the size of the filler **501** may be the smaller or the maximum size possible in a pixel (or sub-pixel) area. In another example, the filler **501** size may be larger than the pixel or sub-pixel area. The filler **501** may have a different or a similar shape as the pixel area on the system substrate **100**. The processes mentioned in FIG. 3 and FIG. 4 may then be applied to improve the light extraction from the micro devices **102a** and **102b**.

(115) Referring to FIG. 6A, in an embodiment where the pixel **601** is comprised of two sub-pixels **601a** and **601b**, the filler **501** is patterned to define the active area of the pixel **601** (active area being defined as the area from which the display emits light). Here, the active area can be smaller, larger, or the same size as the pixel (or sub-pixel) area. As shown in FIG. 6B, FIG. 6C, and FIG. 6D processes mentioned in FIG. 2 and FIG. 3 may be applied. This configuration manages the discoloration at the edges due to the separation between sub-pixels **601a** and **601b**.

(116) Referring to FIG. 6B, a dielectric layer **201** and a reflective layer **202** may be formed around over pixel **601**.

(117) Referring also to FIG. 6C, a black matrix **203** may be formed between adjacent pixels **601** and around each sub-pixel **601a** and **601b** to reduce the reflection of the ambient light.

(118) Referring to FIG. 6D, a transparent conductive layer **301** may be deposited on top of the substrate **100**, covering the black matrix **203** and the top surface of micro LEDs **601a** and **601b**.

(119) In another embodiment shown in FIG. 6E, reflective or other optical components **602** may be formed on the substrate **100** to enhance out-coupling of light produced by the micro device **601a** and **601b**. The common contact **301** is transparent for the light to output through this layer. These structures may be referred to as top emission structures.

(120) Referring to FIG. 6F, the contact pads **101a** and **101b** may be formed to have a concave structure to enhance the out-coupling of light produced by micro-devices **101a** and **101b**. The form of the contact pads **101a** and **101b** is not limited to the concave form and may have other forms depending on the micro device light emission characteristics.

(121) Referring to FIG. 6G, in another embodiment, the structure is designed to output light from

the substrate **100**. In these bottom emission structures, the substrate **100** may be transparent and the common electrode **303** may be comprised a reflective material for better light extraction.

(122) In another embodiment shown in FIG. **6H**, the reflective layer **202** may be extended to cover the micro devices **601a** and **601b**, and act as the common top electrode as well.

(123) In other embodiments, the aforementioned pixel definition structure can cover more than one pixel (or sub-pixel) **601a** and **601b**.

(124) In another case, a reflective layer or the contact pads **101a** and **101b** on the receiving substrate **100** may be used to cover the receiving substrate **100** and create a reflective area before transferring the micro devices **601a** and **601b** for better light out coupling.

(125) In all aforementioned embodiments, the reflective layer can also be opaque. In addition, the reflective layers can be used as one of the electrodes for the micro device **601a** and **601b** or as one of the system substrate connections (electrode, signal, or power line). In another embodiment, the reflective layer can be used as a touch electrode. The reflective layers can be patterned to act as a touch screen electrode. In one case, they can be patterned in vertical and horizontal directions to form the touch screen crossing electrode. In this case, one can use a dielectric between vertical and horizontal traces.

(126) Hybrid Structures

(127) In another embodiment, a thin film electro-optical device **904** is integrated into the receiver substrate **100** after an array of micro devices **801** have been transferred to the receiver substrate **100**.

(128) FIG. **7** illustrates the receiver substrate **100** and lower electrode contact or bonding pads **702a** and **702b** upon which the array of micro devices **801** are transferred, and into which a thin film electro-optical device **904** is integrated in a number of hybrid structure embodiments.

(129) Referring to FIG. **8**, one of the micro devices **801** may be transferred and bonded to the bonding pad **702a** of the receiver substrate **100**. In one case, as shown in FIG. **9**, a dielectric layer **901** may be formed over the receiver substrate **100** to cover the exposed electrodes **702a** and **702b** and any other conductive layers. Lithography and etching may be used to pattern the dielectric layer **901**. A conductive layer **902** is then deposited and patterned to form a bottom electrode of the thin film electro-optical device **904**. If there is no risk of unwanted coupling between the bottom electrode **902** and other conductive layers in the receiver substrate **100**, the dielectric layer **901** may be eliminated. However, the dielectric layer **901** can act as a planarization layer as well to offer better fabrication of the electro-optical devices **904**.

(130) Still referring to FIG. **9**, a bank layer **903** may be deposited on the receiver substrate **100**, e.g. over the dielectric layer **901** and the micro device **801**, to cover the edges of the bottom electrode **902** and the micro device **801**. The thin film electro-optical device **904** may then be formed over the bank layer **903** and the bottom electrode **902** structures. Organic LED (OLED) devices are an example of such a thin film electro-optical device **904**, which may be formed using different techniques, such as but not limited to shadow mask, lithography, and printing patterning. Finally, a top electrode **905** of the electro-optical thin film device **904** is deposited and patterned if needed.

(131) In an embodiment in which the thickness of the micro devices' **801** is significantly high, cracks or other structural problems may occur within the bottom electrode **902**. In these embodiments, a planarization layer **903** may be used in conjunction with or without the dielectric layer **901** to address this issue.

(132) In another embodiment shown in FIG. **10**, the micro device **801** may have an upper device electrode **1001**. The upper device electrode **1001** may be common between other micro devices **801** in or on the system substrate **100**. In this case, the planarization layer **901** (if present) and/or the bank structure **903** covers the upper device electrode **1001** to insulate the upper electrode **1001** from the electro-optical thin film device **904** and the top and bottom electrodes **902** and **905**, to avoid any shorts between the electro-optical device **904** and device electrode **1001**.

(133) Referring to FIG. **11**, in one embodiment, the top electrode **905** of the thin film electro-

optical device **904** may be connected to the micro device **801** through an opening **1005** in the bank (planarization) layer **903** and the electro-optical thin-film device **904**. In this case, the electro-optical thin-film device **904** may be formed selectively so that it is not covering the opening **1005**.

(134) In another case, the lower electrode **702a** of the micro device **801** may be shared between the thin film electro-optical device **904** and the transferred micro device **801**.

(135) Referring to FIG. **12**, in another example, the bottom electrode **902** of the thin film electro-optical device **904** can be extended over the micro device **801**, so that the thin film electro-optical device **904** may be superposed over or around the micro device **801**. In case the micro device **801** needs to have a transparent path to the outside through its top electrode **1001**, the bottom electrode **902** (if not transparent) needs to have an opening over the micro device **801** (for example as shown in FIG. **13A** in association with another embodiment). In this case, the opening can be covered by the bank layer **903** as well. The opening is not limited to the specific structure illustrated in FIG. **12** and can be developed with different methods.

(136) Still referring to FIG. **12**, the micro device **801** may have a transparent path through the substrate **100**, if the lower electrode **702a** is transparent. In a case where a transparent path is required through the upper electrode **1001**, either the bottom electrode **902** and the micro device upper electrode **1001** need to be transparent or there needs to be a combination of openings in one or both of the upper electrode **1001** and the bottom electrode **902** and transparency in one or both of the upper electrode **1001** and the bottom electrode **902**.

(137) FIG. **13A** shows a layout structure where the bottom electrode **902** has an opening **1301** to allow a transparent path through to the top electrode **905**. The opening **1301** may also extend through the bank layer **903** for the common top electrode **905**. If there is no common top electrode **905** and if the bank layer **903** is transparent, the opening **1301** in the bank layer **903** is not needed. In some embodiments, if the top electrode **905** is also opaque, the opening **1301** in the top electrode **905** is also needed for top emission.

(138) Referring to FIG. **13B**, in another embodiment, to provide a transparent path for the micro device **801**, the bottom electrode **902** does not cover the micro device **801**. There can be an opening **1301** in the bank layer **903** for a common top electrode **905**. If there is no common top electrode **905** and the bank layer **903** is transparent, the opening **1301** in the bank layer **903** is not needed.

(139) In another case, the contact pad structures **702b** of the thin film electro-optical device **904** can be extended to act as reflective layer. As can be seen in FIG. **14A**, the two side-by-side pixels with contacts **702b** can act to laterally confine the light generated by the micro device **801** in the pixel. In another embodiment shown in FIG. **14B**, a reflective layer **1401** mounted on a top surface of the substrate **100** can reflect more of the light toward the top electrode **905**. As a result, the out coupling of the light generated by the micro device **801** is enhanced. In this case, the best practice is either to make both top and bottom electrodes **902** and **905** of the thin film electro-optical device **904** transparent, or make openings if the electrodes **902** and **905** are opaque.

(140) In another embodiment, the thin film electro-optical devices **904** and micro devices **801** may be on two opposite sides of the system substrate **100**. In this case, the system substrate circuitry can either be on one side of the system substrate **100** and connected to the other side through contact holes or, the circuits can be on both sides of the system substrate **100**.

(141) In another case, the micro device **801** may be on one system substrate **100** and the thin film electro-optical device **904** on another system substrate. These two substrates may then be bonded together. In this case, the circuit can be on one of the system substrates or on both substrates.

(142) FIGS. **14C** and **14D** show different structures in which the micro device (LED) **801**, and the thin film electro-optical device **904** are integrated to develop a semiconductor device. Here a reflective or light confinement structure **5601** extending along the substrate **100** beneath both the micro-device **801** and the thin film electro-optical device **904**, ideally mounted on the substrate **100**, is used to direct the light output of the micro device **801**. The reflective structure **5601** may be the same as the micro device electrode **702a** or a separate electrode **702a** may be used. As shown in

FIG. 14D, a first part **902a** of the bottom electrode **902**, which extends from the contact pad **702b** to a main planar section **902b** parallel to the optoelectronic device **904** may be comprised of a reflective material capable of being used as a light confinement or reflective structure for directing light in the desired direction, e.g. through the substrate **100** or top electrode **905**, and preventing light from entering adjacent pixels. The first part **902a** of the bottom electrode **902** can be deposited first followed by the rest of the bottom electrode **902b** or the first part **902a** can be deposited after the main planar section **902b**. Also, other electrodes can be deposited to connect the micro device **801**. Ideally, both bottom and top electrodes **902** and **905** of the thin film optoelectronic device **904** are transparent, so that the light from the micro device **801** and the optoelectronic device **904** are emitted through the top electrode **905**. Any stray light not emitted outwardly may be redirected by the reflective structure **5601** and **902a** outwardly through the top electrode **905**. The optoelectronic device **904** can be formed between bank structure **903**, which may alternatively be black matrix. After the optoelectronic device **904**, other structures can be integrated, such as encapsulation.

(143) FIGS. 14E, 14F and 14G describe another structure in which the optoelectronic device **904** and the micro device **801** are side by side on the substrate **100**. Here, the micro device **801** is transferred to the system substrate **100**. A planarizing layer **903** (or bank layer) is deposited and patterned to open an area for the optoelectronic device **904**. The optoelectronic device **904** is deposited with different possible methods, such as vapor deposition, printing, and so on. Then top electrode **905** is deposited over top both the optoelectronic device **904** and the micro device **801**. Here, other structures can be deposited after the top electrode **905**. The light can go through the top electrode **905** or the system substrate **100**. In one structure described in FIG. 14E, the micro device **801** and the optoelectronic device **905** have the same top electrode **905**. In another structure described in FIG. 14F, the micro device **801** has a separate upper electrode **5620** covered by the passivation (dielectric) layer **903**. Here, the top electrode **905**, which extends over both the thin film electro-optical device **904** and the micro device **801**, may be used as a light confinement/reflective structure to guide the light from the micro device **801** and the thin film electro-optical device **904** through the system substrate **100**. In another example shown in FIG. 14G, the top electrode **905** is shared; however, the passivation/planarization layer **903** is patterned at **5622** into one or more concave structures to create one or more light confinement/reflective structures for directing light in the desired direction, e.g. back through the substrate **100**. Here a separate layer can be used to create the light confinement structure before depositing the top electrode **905**.

(144) In all structures described here, the contact electrode(s) **702a** for the micro device **801** can be deposited after the micro device **801** is transferred to the system substrate **100** or a contact **702a** can pre-exist before the transfer process. There can be a planarization layer **901** or **903** before the micro device **801** and other devices are mounted on the system substrate **100** to improve surface profile. In this case, there can be openings to connect the micro device **801** to the other elements in the system substrate **100**.

(145) Structures described hereinbefore can be combined. For example, the light confinement or extraction structures and hybrid devices can be mixed.

(146) In hybrid devices described here in FIGS. 9 to 14, thin film light emitting device structures **904** can include a color filter. In the case of using color filters, the light from the micro device **801** needs to pass the thin film light emitting device structure **904** and through the color filter.

(147) FIGS. 14H and 14I show two example of integrating micro device **801** with a color filter **5810** and the optoelectronic device **904**. In this case, the light goes through the system substrate **100**. After deposition of the color filter **5810**, location of the bonding pads **702a** and **702b**, and transfer of the micro device **801** and the upper electrode **1001** (the order of these steps may be changed), a planarization, e.g. dielectric, layer **901** may be deposited, if required. Two or three different planarization layers may be deposited after each previous step. Then the optoelectronic

device **904** and the bottom and top electrodes **902** and **905**, respectively, are deposited and formed. Here, the micro device **801** may include a light confinement structure before transfer (this can be used for other structures in this document as well). Also, as shown in FIG. **14I**, the light confinement structure can be formed after the transfer of the micro device **801**. In the illustrated embodiment, the light confinement has a passivation layer **5814**, thereby insulated from the micro device **801** and upper electrode **1001**, and a reflective layer **5812** formed into a concave structure for reflecting light in a desired direction, e.g. through the substrate **100**.

(148) FIG. **14J** shows another example of integrating the micro device **801** with the optoelectronic device **904** and the color filter **5810** mounted over the optoelectronic device **904** and the top electrode **905**. All the above structures, such as light confinement structures, can be used with micro device **801** and the optoelectronic device **904** in this example. Here, the light goes through the transparent top electrode **905**. There may be a transparent protective layer **5910** (or layers) in between the top electrode **905** and the color filter **5810** in this structure.

(149) Integration

(150) This document also discloses various methods for the integration of a monolithic array of micro devices into a system substrate or selective transferring of an array of micro devices to a system substrate. Here, the proposed processes are divided into two categories. In the first category, the pitch of the bonding pads on the system substrate is the same as the pitch of the bonding pads of the micro devices. In the second category, bonding pads on the system substrate have a larger pitch compared to that of the micro devices. For the first category, three different schemes of integration or transfer are presented 1. Front-side Bonding 2. Back-side bonding 3. Substrate Through via Bonding.

(151) In this embodiment micro-devices may be of the same type or different types in terms of functionality. In one embodiment, micro-devices are micro-LEDs of the same color or of a number of different colors (e.g. Red, Green, and Blue), and the system substrate is the backplane, controlling individual micro-LEDs. Such multi-color LED arrays are fabricated directly on a substrate or transferred to a temporary substrate from the growth substrate. In one example shown in FIG. **15**, RGB micro-LED devices **1503**, **1504**, and **1505** were grown on a sacrificial/buffer layer **1502** and the substrate **1501**. In one case, the system substrate **1506** having contact pads **1507** can be aligned (FIG. **16**) and bonded to the micro device substrate **1501** as it is shown in FIG. **17**. After removing the micro-device substrate **1501** (FIG. **18**) and sacrificial/buffer layer **1502** (FIG. **19**), a filler dielectric coating **2001** (e.g. Polyimide resist) may be spin-coated/deposited on the integrated sample (FIG. **20**). This step may be followed by an etching process to reveal the tops of the micro-LED devices. In the case of micro-LED devices, a common transparent electrode **2002** may be deposited on the sample. In another embodiment, a top electrode may be deposited and patterned to isolate micro devices for subsequent processes.

(152) In another embodiment, as shown in FIG. **21** the micro devices **1503**, **1504**, and **1505** are grown on a buffer/sacrificial layer **1502**. A dielectric filler layer **2101** is deposited/spin-coated on the substrate to fully cover the micro devices. In one example illustrated in FIG. **21**, this step is followed by an etching process to reveal the tops of the micro devices **1503**, **1504**, and **1505** to form the top common contact and seeding layer for subsequent processes (e.g. electroplating). Referring to FIG. **22**, a thick mechanical supporting layer **2102** is then deposited, grown or bonded on the tops of the sample. Here, the filler layer **2101** can be a black matrix layer or a reflective material. Also, before depositing the mechanical support, one can deposit an electrode (either as a patterned or a common layer). The mechanical support layer is then deposited. In the case of optoelectronic devices such as LEDs, the mechanical support layer needs to be transparent. As shown in FIG. **23** and FIG. **24**, the micro device substrate **1501** is then removed using various processes such as laser lift-off or etching. In one case, the thickness of the substrate is initially reduced to a few micrometers by processes such as but not limited to deep reactive ion etching (DRIE). The remaining substrate then is removed by processes such as but not limited to a wet

chemical etching process. In this case, the buffer/sacrificial layer **1502** may act as an etch-stop layer to ensure a uniform etched sub-surface and to avoid any damage to the micro devices. After removing the buffer layer **1502** as shown in FIG. **24**, another etching (e.g. ME) is performed to expose the micro devices. One may deposit and pattern a metallic layer to serve as the upper contact and bond pads for the micro-devices if they haven't been formed during the micro device fabrication. The system substrate **1506** having contact pads **1507** can then be aligned and bonded to the micro device array as shown in FIG. **25**. Depending on the type and functionality of the micro devices the mechanical supporting layer **2102** and filler layer **2101** may be then removed as shown in FIG. **26A** and FIG. **26B**.

(153) In another embodiment, through substrate vias are implemented to make contacts to the back of the micro devices.

(154) Referring to FIG. **27**, in one embodiment, the micro devices **1503**, **1504**, and **1505** may be multicolor micro-LEDs grown on an insulating buffer layer **1502**. This buffer layer **1502** may function as an etch-stop layer as well. A dielectric layer **2701** is deposited as a filler layer over and around the micro devices **1503**, **1504** and **1505**.

(155) Referring to FIG. **28A** and FIG. **28B**, using processes such as but not limited to photolithography, patterns are formed on the backside of the substrate **1501**. In one embodiment, a method such as DRIE is used to make through substrate holes in the substrate **1501**. Buffer layer **1502** which may act as an etchstop layer may be removed using for example a wet-etch process.

(156) Referring to FIG. **29**, an insulating film **2901** may be deposited on the back of the substrate **1501**. The insulating layer **2901** may be partially removed from back side of the micro devices **1503**, **1504**, and **1505** to allow formation of electrical contacts to these micro devices.

(157) Referring to FIG. **30**, the through holes are filled with a conductive material **3001** using processes such as but not limited to electroplating. Here, the vias may act as the micro device contacts and bonding pads.

(158) As illustrated in FIG. **31**, a common front contact **3101** of the micro devices **1503**, **1504**, and **1505** is formed by performing an etching process (e.g. using ME) to reveal the tops of the micro devices **1503**, **1504** and **1505** followed by the deposition of a transparent conductive layer to form the front contact **3101**.

(159) Referring to FIG. **32**, the micro device substrate **1501** is then aligned and bonded to the system substrate **1506** having contact pads **1507**, which in this example may be a backplane controlling individual devices.

(160) In another embodiment, micro devices have been fabricated on a substrate with arbitrary pitch length to maximize the production yield. For example the micro devices may be multi-color micro-LEDs (e.g. RGB). The system substrate for this example may be a display backplane with contact pads having a pitch length different than those of the micro-LEDs.

(161) Referring to FIG. **33A**, in one embodiment, the donor substrate **1501** has micro device types **3301**, **3302**, and **3303** and they are patterned in the form of one dimensional arrays **3304** in which for each micro device **3301**, **3302**, or **3303** from one type, there is at least a micro device from another type that their pitch **3305** matches the pitch of the corresponding areas (or pads) on the receiver (or system) substrate **1506**.

(162) As an example, in one embodiment shown in FIG. **33B**, the pitch **3404** of contact pads **1507** is two times larger than the pitch **3402** of the micro devices **3401** as shown in FIG. **33**.

(163) Referring to FIG. **34**, the system substrate **1506** and micro device substrate **1501** are brought together, aligned and put in contact.

(164) As shown in FIG. **35** and FIG. **36**, methods such as laser lift-off (LLO) may be used to selectively transfer the micro devices **3401** to the contact pads **3403** on the system substrate **1506**. As shown in FIG. **37**, transfer may be followed by depositing a filler layer **3701** and a conformal conductive layer **3702** on top of the system substrate as the common electrode.

(165) In another embodiment shown in FIGS. **38A** and **38B**, a buffer layer **3801** is necessary as a



material template for the fabrication of micro devices **1503**, **1504**, and **1505**.

(166) Still referring to FIG. **38A** and FIG. **38B**, the buffer layer **3801** is deposited on the sacrificial layer **1502** and patterned to isolate micro devices **1503**, **1504**, and **1505**. In some cases, the sacrificial layer **1502** may be patterned as well.

(167) In one embodiment, instead of isolating individual micro devices, groups of micro devices may be isolated from one another (as shown in FIG. **38A** and FIG. **38B**) to facilitate the transfer process.

(168) Referring to FIG. **39**, a filling material **3901**, such as but not limited to polyimide, may be spin coated on the substrate **1501** to fill the gap between the individual micro devices **1503**, **1504** and **1505**. This filling step insures the mechanical strength during the transfer process. This is particularly important when a process like laser lift-off is used to detach micro devices from the carrier substrate.

(169) Referring to FIG. **40**, the micro devices may not have the same height which make it difficult to bond them to the system substrate **1506**. In these cases, one can implement an electrostatic grip mechanism **4001** or other grip mechanisms in the system substrate **1506** to temporarily keep the micro devices on the system substrate **1506** for the final bonding steps. The grip mechanism **4001** may be local for micro devices or a global grip for a group of micro devices as in the case of same-pitch transfer for the whole wafer. The grip mechanism **4001** may be on a layer above the contact electrode **1507**. In this case, a planarization layer may be used.

(170) In one embodiment, referring to FIG. **41A**, the pattern of different micro device types **3301**, **3302**, and **3303** on donor substrate create a two dimensional array of each type (for example array **4100**) where the pitch between the arrays **4101** defined as the center-to-center distance between adjacent arrays matches the pitch of the corresponding area on the system substrate.

(171) In one embodiment shown in FIG. **41B** and FIG. **42**, when sub-device pitch **4103** is larger than the normal distance between fabricated individual micro devices **1503** on their substrate (e.g. in large displays), micro device substrate **1501** is laid out in the form of two-dimensional single color arrays. Here, the pitch **4102** of the contact pads **1507** and the pitch **4103** of the micro device array **1503** are the same. Using this technique, one may relax the micro device fabrication requirements and reduce the selective transferring process as compared to that described above.

(172) FIG. **43** and FIG. **44** shows an alternative pattern where micro devices **1503** are not formed in two-dimensional groups, and in which the different micro devices **1503** are uniformly placed across the substrate **1501** as it shown in FIG. **43** for three different micro devices **1503**.

(173) Referring to FIG. **45**, in another embodiment, micro devices **4503** are first transferred to a conductive semi-transparent common substrate **4501**, then they are bonded to a system substrate **4502**.

(174) Color Conversion Structure

(175) In some embodiments where the micro devices are optical devices such as LEDs, one can use either color conversion or color filters to define different functionality (different colors in the case of pixels). In this embodiment, two or more contact pads on the system substrate are populated with the same type of optical device. Once in place, the devices on the system substrates are then differentiated by different color conversion layers.

(176) Referring to FIG. **46A** and FIG. **46B**, in one embodiment, after transferring micro devices **1503** to the system substrate **1506**, the whole structure is covered by a planarization layer **4601**. A common electrode **4602** is then formed on the planarization layer **4601**. The planarization layer can be the same height as, taller than, or shorter than the stacked devices. If the planarization layer **4601** is shorter (or there is no planarization layer) the wall of the device can be conformally covered by passivation materials.

(177) Referring to FIG. **47**, a bank structure **4701** is developed (especially if a printing process is used to deposit the color conversion layers). The bank **4701** can separate each pixel or just separate different color conversion materials **4702**.

(178) FIG. **48** shows an integrated structure where the color conversion layer **4702** fully covers the top of the transferred micro devices and partially covers their sides. Bank **4701** separates the color conversion layers **4702** and the electrode **4602** is a common contact for all transferred micro devices.

(179) FIG. **49** shows an integrated structure where the color conversion layer **4702** fully covers the top of the transferred micro devices and partially covers their sides. Bank **4701** separates the color conversion layers **4702** and contacts **1507** to the micro devices are made only through the system substrate **1506**.

(180) FIG. **50** shows an integrated structure where the color conversion layer **4702** is directly formed on the common electrode **4602**. In this case no bank layer is used.

(181) FIG. **51** shows an integrated structure where the color conversion layer **4702** fully covers the top of the transferred micro devices and partially covers their sides. The electrode **4602** is a common contact for all transferred micro devices. In this case no bank layer is used.

(182) FIG. **52** shows an integrated structure where the color conversion layer **4702** fully covers the top of the transferred micro devices and partially covers their sides. The contacts to the micro devices are made only through the system substrate **1506**. In this case no bank layer is used.

(183) In one embodiment, shown in FIG. **53A** and FIG. **53B**, after forming the color conversion material **4702** on the integrated system substrate **1506**, a planarization layer **5301** is deposited on the structure. In some cases where the color conversion material and/or other components of the integrated substrate need to be protected from environmental conditions, an encapsulation layer **5302** is formed over the whole structure. It should be noted that the encapsulation layer **5302** may be formed from a stack of different layers **5302-2** to effectively protect the integrated substrate from environmental conditions.

(184) Referring to FIG. **54A** and FIG. **54B**, in another embodiment a separate substrate **5401** coated with the encapsulation layer **5302** may be bonded to the integrated system substrate.

(185) The embodiments depicted in FIG. **53** and FIG. **54** may be combined in which encapsulation layer **5302** is formed both on the structure **1506** and the separate substrate **5401** for more effective capsulation.

(186) The common electrode is a transparent conductive layer deposited on the substrate in the form of a blanket. In one embodiment, this layer can act a planarization layer. In some embodiments, the thickness of this layer is chosen to satisfy both optical and electrical requirements.

(187) The distance between the optical devices may be chosen to be large enough so as to reduce cross-talk between the optical devices or a blocking layer is deposited between the optical devices to achieve this. In one case, the planarization layer functions also as a blocking layer.

(188) After the color conversion layers are deposited, different layers such as polarizers can be deposited.

(189) In another aspect, color filters **4702-1** are deposited on the color conversion layers. In this case wider color gamut and higher efficiency may be achieved. One can use a planarization layer and/or bank layer after the color conversion layer before depositing the color filter layers.

(190) The color filters can be larger than the color conversion layer to block any light leakage. Moreover, a black matrix **4702-2** can be formed between the color conversion islands or color filters.

(191) FIGS. **55A**, **55B**, and **55C** illustrates structures where the device is shared between a few pixels (or sub-pixels). Here the micro device **1503** is not fully patterned but the horizontal condition is engineered so that the contacts **1507** define the area allocated to each pixel. FIG. **55A** shows the system substrate **1506** with contact pads **1507** and a donor substrate **1501** with micro devices **1503**. After the micro devices **1503** are transferred to system substrate (shown in FIG. **55B**), one can do post processing (FIG. **55C**) such as depositing common electrode **4602**, color conversion layers **4702**, color filters and so on. FIG. **55C** shows one example of depositing color

conversion layers **4702** on top of the micro device **1503**. Here, the color conversion layer can be followed by color filter deposition. The methods described in this disclosure and/or other possible method can be used for different parts or integration of different layers into the display. Also, one can use planarization layer **4601** before or after electrode **4602**. Also a reflective layer **1509** can be used between the LEDs **1503**. Filler can be used between the pads **1507** and the filler can be a black matrix. Some spacer or light confinement structure can be used between LEDs **1503**. On the system substrate **1506** a reflective layer can be used to direct the light.

(192) FIG. **55D** shows another example of depositing color conversion layers **4702** on top of the micro device **1503**. However, the other methods described in this disclosure and other possible methods can also be used. Also, one can use planarization layer **4601** before or after transparent upper electrode **4602**. Also a reflective layer can be used between the LEDs **1503**. Filler can be used between the pads **1507** and the filler **5502** can be a black matrix. Some spacer or light confinement structure **5504** can be used between micro-LEDs **1503**. On the system substrate **1506** a reflective layer **1509** can be used to direct the light. Here, the light of micro-LEDs **1503** is spread over larger area so that there is less stress on the color conversion layer **4702**. Different structure can be used to spread the light over larger area as well.

(193) It is possible to add the color conversion layers as described into pixel (or sub-pixel) active areas after formation of the active area. This can offer a higher fill factor and higher performance and also avoid color leaking from the side pixel (or sub-pixel) if the active area of the pixel (or sub-pixel) is covered by reflective layers.

(194) The foregoing description of one or more embodiments of the invention has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. It is intended that the scope of the invention be limited not by this detailed description, but rather by the claims appended hereto.

## Claims

1. A method to develop color pixels using micro-LEDs with pixels, the method comprising: having a system substrate with contact pads and a donor substrate with the micro-LEDs; transferring the micro-LEDs to the system substrate; forming bank structures isolating light from each of the micro-LEDs; and depositing color conversion layers on a top of the micro-LEDs within the bank structures, wherein the micro-LEDs are shared between at least two pixels or sub-pixels and each contact pad defines an area for the at least two pixels or sub-pixels.
2. The method of claim 1, further comprising providing a spacer or light confinement structure between the micro-LEDs.
3. The method of claim 1, wherein a horizontal contact is engineered so that the horizontal contact defines the area allocated to the at least two pixels or sub-pixels.
4. The method of claim 1, wherein a planarization layer is used before or after a transparent upper electrode.
5. The method of claim 1, wherein a filler is used between the contact pads.
6. The method of claim 5, wherein the filler is black matrix.
7. The method of claim 1, further comprising using a reflective layer to direct the light on the system substrate.
8. The method of claim 1, wherein the color conversion layers are deposited into pixel or sub-pixel active areas after formation of an active area.
9. A method to develop color pixels using micro-LEDs with pixels, the method comprising: having a system substrate with contact pads and a donor substrate with the micro-LEDs; transferring the micro-LEDs to the system substrate; forming bank structures isolating light from each of the micro-LEDs; and depositing color conversion layers on a top of the micro-LEDs within the bank

- structures, wherein a filler is used between the contact pads, and wherein the filler is black matrix.
10. The method of claim 9, further comprising providing a spacer or light confinement structure between the micro-LEDs.
11. The method of claim 9, wherein the micro-LEDs are shared between at least two pixels or sub-pixels and each contact pad defines an area for the at least two pixels or sub-pixels, and wherein a horizontal contact is engineered so that the horizontal contact defines the area allocated to the at least two pixels or sub-pixels.
12. The method of claim 9, wherein a planarization layer is used before or after a transparent upper electrode.
13. The method of claim 9, wherein the micro-LEDs are shared between at least two pixels or sub-pixels and each contact pad defines an area for the at least two pixels or sub-pixels, and the method further comprising using a reflective layer to direct the light on the system substrate.
14. The method of claim 9, wherein the color conversion layers are deposited into pixel or sub-pixel active areas after formation of an active area.
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